

Ex No: 02	Scheduling of Project with PERT and CPM Techniques
Date:	

Aim

To estimate the project completion time by creating a project schedule, defining task dependencies, generating the CPM and PERT network diagrams, assigning resources and costs, and calculating the overall project duration using ProjectLibre

Algorithm

1. Open ProjectLibre.
2. Create a new project.
3. Enter project name and start date.
4. Create project activities with durations.
5. Define predecessor relationships.
6. Generate Gantt Chart.
7. Generate Network Diagram (PERT/CPM).
8. Create Resource Sheet.
9. Assign resource costs.
10. Allocate resources.
11. View Project Summary and Cost Report.
12. Save the project

Procedure

Step 1: Create a New Project

1. Open **ProjectLibre**.
2. Click **File** → **New**.
3. Enter:
 - **Project Name:** Software Project Scheduling
 - **Manager:** Your Name

- **Start Date:** Select today's date.

4. Click **OK**.

Step 2: Enter Project Activities

In the **Gantt Chart**, enter the following activities:

Task Name	Duration
Requirement Gathering	2d
Feasibility Analysis	3d
UI/UX Design	4d
Database Development	5d
Application Development	6d
System Testing	3d
Bug Resolution	2d
Project Deployment	1d

Step 3: Enter Predecessors

In the **Predecessors** column, enter:

Task	Predecessor
Requirement Gathering	-
Feasibility Analysis	1
UI/UX Design	2

Task	Predecessor
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Database Development	3
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Application Development	4
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System Testing	5
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Bug Resolution	6
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Project Deployment	7
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ProjectLibre will automatically schedule the project.

Step 4: View the Gantt Chart

After entering the tasks and predecessors:

- The **Gantt Chart** will be generated automatically on the right side.
- This shows the start date, finish date, and duration of each activity.

Step 5: View the Network Diagram (PERT/CPM)

1. Click **View**.
2. Select **Network Diagram**.

ProjectLibre automatically creates the **PERT/CPM Network Diagram** showing the sequence of project activities.

Step 6: Create Resources

Go to **View → Resource Sheet** and enter:

Resource Name Standard Rate

Arun Kumar	₹2000/day
Priya Sharma	₹1800/day
Karthik Raj	₹2500/day
Nisha Devi	₹2200/day
Rahul Verma	₹2700/day
Deepika S	₹2100/day
Ajay Kumar	₹1900/day
Meena R	₹2300/day

Step 7: Assign Resources

Return to the **Gantt Chart**.

Assign the resources:

Task	Resource
Requirement Gathering	Arun Kumar
Feasibility Analysis	Priya Sharma
UI/UX Design	Karthik Raj
Database Development	Nisha Devi
Application Development	Rahul Verma
System Testing	Deepika S
Project Deployment	Meena R

Task	Resource
Bug Resolution	Ajay Kumar

Step 8: View Project Summary

After assigning resources:

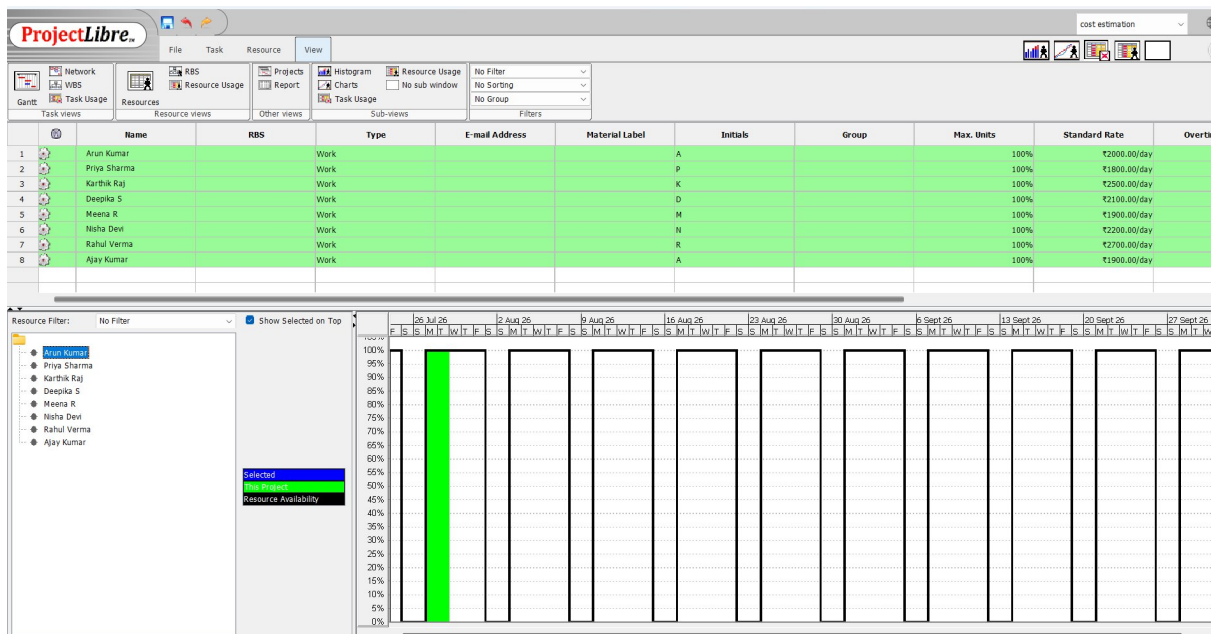
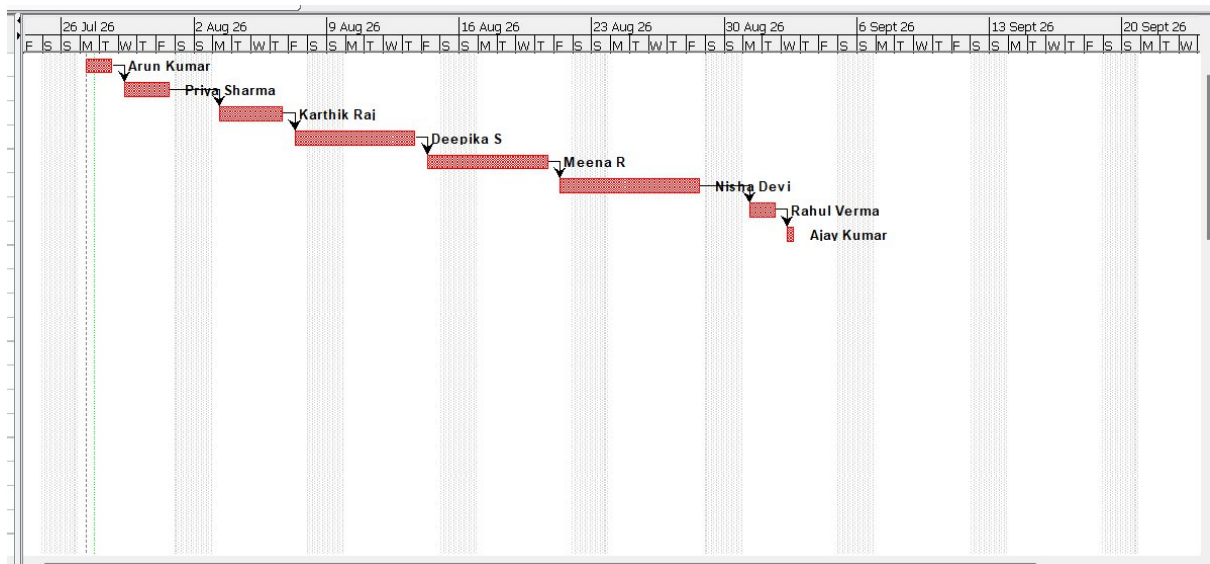
- Click **Project** or **Reports** (depending on your ProjectLibre version).
- View:
 - Total Duration
 - Total Cost
 - Resource Allocation

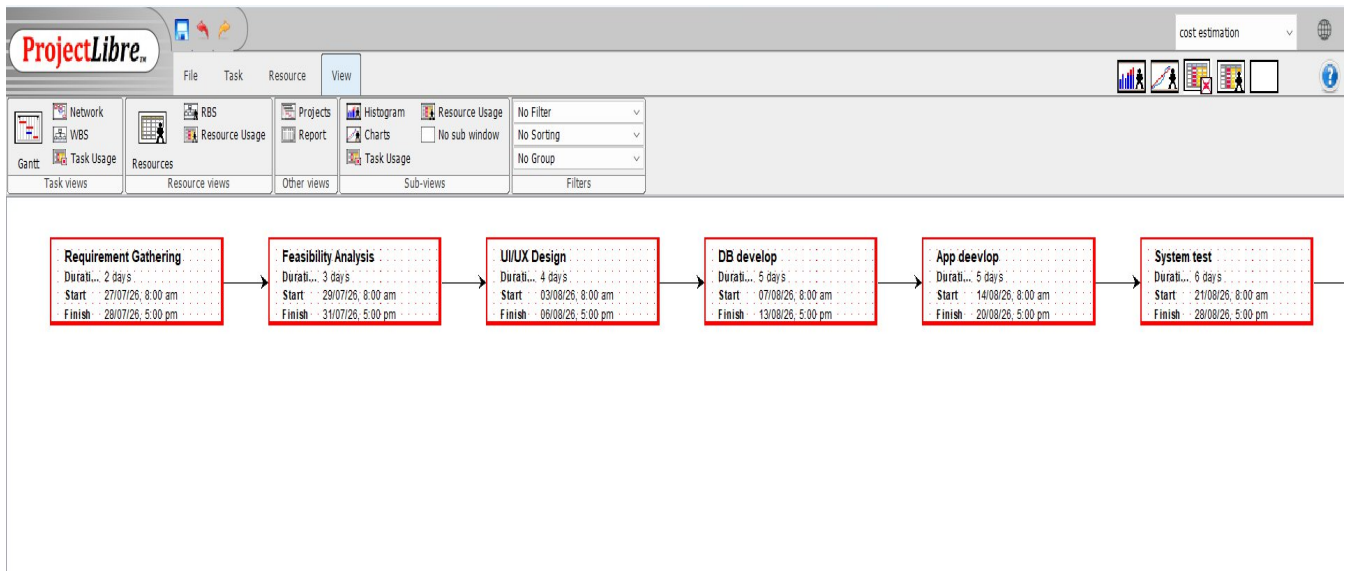
Observation:

Project Duration: 26 Days 208 Hours

Critical Path:

Requirement Gathering → Feasibility Analysis → UI/UX Design → Database Development
→ Application Development → System Testing → Bug Resolution → Project Deployment





Name	Start	Finish	Manager	Status Date	Cost	Work	Actual Cost	Actual Work	Notes	Forward scheduled
Expt 1	27/07/26, 8:00 am	27/07/26, 8:00 am		27/07/26, 5:00 pm	₹0.00	0 hours	₹0.00	0 hours		
cost estimation	27/07/26, 8:00 am	02/09/26, 5:00 pm	Nalini	27/07/26, 5:00 pm	₹59900.00	224 hours	₹0.00	0 hours		

Result:

The project schedule was successfully created using **PERT and CPM techniques** in ProjectLibre. The software automatically generated the **Gantt Chart** and **Network Diagram**, identified the **critical path**, and estimated the **project completion time of 26 days (208 hours)**.